Assignment 1 – File Permissions Using chmod

1. Create a file named student.txt.

```
vaibhavi@vaibhavi99099W1:~$ touch student.txt
vaibhavi@vaibhavi99099W1:~$ ls
Desktop    my_directory  my_new_file.txt    permission_dir  test_directory
backup     my_file.txt   my_old_files       sample.txt      text.txt
banks      my_files      my_other_file.txt  student.txt
blank.txt  my_music      my_third_file.txt  test.txt
vaibhavi@vaibhavi99099W1:~$
```

2. Check the current permissions of student.txt.

```
~$ ls -l student.txt
-rw-r--r-- 1 vaibhavi 0 Oct 27 10:59 student.txt
~$
```

3. Set the permissions of student.txt to 755 using numeric notation.

```
vaibhavi@vaibhavi99099W1:~$ chmod 755 student.txt
vaibhavi@vaibhavi99099W1:~$ ls -l student.txt
-rwxr-xr-x 1 vaibhavi vaibhavi 0 Oct 27 10:59 student.txt
vaibhavi@vaibhavi99099W1:~$
```

4. Set the permissions of student.txt to 644 using numeric notation.

```
vaibhavi@vaibhavi99099W1:~$ ls -l student.txt
-rw-r--r-- 1 vaibhavi   0 Oct 27 10:59 student.txt
vaibhavi@vaibhavi99099W1:~$
```

5. Using symbolic notation:
   Add execute permission for the owner.

```
vaibhavi@vaibhavi99099W1:~$ chmod u+x student.txt
vaibhavi@vaibhavi99099W1:~$ ls -l student.txt
-rwxr--r-- 1 vaibhavi vaibhavi 0 Oct 27 10:59 student.txt
vaibhavi@vaibhavi99099W1:~$
```

   Remove write permission from the group.

```
~$ chmod g-w student.txt
~$ ls -l student.txt
-rwxr--r-- 1 vaibhavi 0 Oct 27 10:59 student.txt
~$
```

   Remove all permissions from others.

```
vaibhavi@vaibhavi99099W1:~$ chmod o-rwx student.txt
vaibhavi@vaibhavi99099W1:~$ ls -l student.txt
-rwxr----- 1 vaibhavi 0 Oct 27 10:59 student.txt
vaibhavi@vaibhavi99099W1:~$
```

6. Set student.txt permissions so that: Owner: Read, Write, Execute Group: Read, Execute Others: Read

```
vaibhavi@vaibhavi99099W1:~$ chmod o-rwx student.txt
vaibhavi@vaibhavi99099W1:~$ ls -l student.txt
-rwxr----- 1 vaibhavi 0 Oct 27 10:59 student.txt
vaibhavi@vaibhavi99099W1:~$
```

7. Create a directory named project.

```
vaibhavi@vaibhavi99099W1:~$ mkdir project
vaibhavi@vaibhavi99099W1:~$ ls -ld project
drwxr-xr-x 2 vaibhavi 4096 Oct 27 11:18 project
vaibhavi@vaibhavi99099W1:~$
```

8. Set the permission of project to 750.

```
vaibhavi@vaibhavi99099W1:~$ mkdir project
vaibhavi@vaibhavi99099W1:~$ ls -ld project
drwxr-x--- 2 vaibhavi 4096 Oct 27 11:18 project
vaibhavi@vaibhavi99099W1:~$
```

9. Create a file inside project and apply the required permission using symbolic notation.

```
vaibhavi@vaibhavi99099W1:~$ touch project/document.txt
vaibhavi@vaibhavi99099W1:~$ chmod u+rwx project/document.txt
vaibhavi@vaibhavi99099W1:~$ chmod g+r project/document.txt
vaibhavi@vaibhavi99099W1:~$ chmod o-rwx project/document.txt
vaibhavi@vaibhavi99099W1:~$ ls -l project/document.txt
-rw-r----- 1 vaibhavi vaibhavi 0 Oct 27 11:21 project/document.txt
vaibhavi@vaibhavi99099W1:~$
```

10. Verify all permission changes using ls -l.

```
vaibhavi@vaibhavi990999W1:~$ ls -l student.txt
-rwxr-xr-- 1 vaibhavi vaibhavi 0 Oct 27 10:59 student.txt
vaibhavi@vaibhavi99099W1:~$ ls -l project
drwxr-x--- 2 vaibhavi vaibhavi 4096 Oct 27 11:21 project
vaibhavi@vaibhavi990999W1:~$ ls -l project/document.txt
-rw-r----- 1 vaibhavi vaibhavi 0 Oct 27 11:21 project/document.txt
vaibhavi@vaibhavi99099W1:~$ l s -l project
total 0
-rw-r----- 1 vaibhavi vaibhavi 0 Oct 27 11:21 document.txt
vaibhavi@vaibhavi99099W1:~$ ls -l project
project:
total 0
-rw-r----- 1 vaibhavi vaibhavi 0 Oct 27 11:21 document.txt
vaibhavi@vaibhavi99099W1:~$
```